Alpha^2: Evidence-Safe Skill Evolution for Reliable Factor Libraries

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Abstract

Automated alpha-mining systems increasingly use large language models and self-evolving agents to generate, evaluate, and refine formulaic predictors. This adaptivity creates a structural validity problem: a nominally held-out window becomes training signal after repeated inspection, so the best validation alpha can improve with search budget even when no predictive signal exists. Chronological splits and heuristic verification do not control false discoveries under adaptive candidate generation, optional continuation, and unbounded search. We introduce Alpha^2—financial alpha under statistical-alpha control—as a proof-carrying framework that lets an agent learn from accumulated experience without reusing accumulated evidence. Each candidate is registered with an immutable formula, birth time, utility contract, transaction-cost rule, and error budget. An evidence firewall allows skills and failure memories to evolve from past observations, but requires every new factor to begin a fresh e-process on post-birth market data. We show that the resulting evidence is anytime-valid for each adaptively generated factor and that predictable alpha spending controls family-wise error across an unbounded stream of candidates. A bounded log-evidence signal then credits skill selection, utilization, and distillation without changing the underlying certification rule. We design a three-layer evaluation on synthetic null markets, planted-signal markets, and chronological equity panels. The completed evaluation will test whether Alpha^2 controls false certification while retaining competitive locked-test RankIC and net portfolio performance.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → *Intelligent agents*; • **Mathematics of computing** → *Sequential analysis*.

Keywords

alpha mining, self-evolving agents, e-processes, adaptive data analysis, multiple testing, quantitative finance

ACM Reference Format:
Anonymous Author(s). 2026. Alpha^2: Evidence-Safe Skill Evolution for Reliable Factor Libraries. In *ACM KDD 2027, August 1–5, 2027, San Jose, CA, USA.* ACM, New York, NY, USA, 9 pages.

1 Introduction

Formulaic alpha mining searches for compact programs that map historical market observations to predictors of future returns. The task is attractive because the resulting expressions are executable, inspectable, and easier to stress-test than opaque end-to-end forecasters. Recent systems have greatly expanded the search loop. Reinforcement-learning methods optimize collections of complementary factors [10, 21]; LLM agents translate economic hypotheses into expressions and refine them from backtest feedback [4, 5, 13]; and newer systems organize experience as factor graphs, skills, and reusable failure memories [2, 11, 16]. These developments increase the number and semantic diversity of hypotheses that can be tested with little human effort.

The same capability changes the statistical nature of the task. An agent does not generate candidates independently of evaluation data. It observes which ideas worked, selects promising parents, rewrites formulas, tunes parameters, and decides when to stop. Consequently, a validation window becomes part of the adaptive search state. With enough iterations, the best validation score can rise even in a market with no exploitable signal. The familiar factor-zoo problem [3] is therefore amplified from human-scale specification search to machine-scale adaptive discovery. Recent work has questioned whether reported returns from LLM trading agents constitute deployment evidence [20]; however, current alpha-mining systems still mainly optimize predictive performance, diversity, or heuristic robustness gates.

Standard safeguards only partially address this failure. A fixed train–test split protects the test window once, but the protection disappears after the agent repeatedly inspects it. Multiple-comparison corrections assume a known or fixed family, whereas an evolving agent creates hypotheses at data-dependent times. Walk-forward evaluation respects chronology but does not by itself control optional stopping: a system may keep observing a candidate until its score crosses a desired threshold. Verifier-guided strategy optimization can reject fragile edits [6], yet hard thresholds do not quantify the probability of false promotion across a growing search history.

This paper asks a sharper question:

> *Can a self-evolving alpha-mining agent learn from sequential market feedback while preserving valid evidence for the factors it discovers?*

We answer this question with Alpha^2, an evidence-safe skill-evolution framework. Its central design principle is to separate *knowledge* from *evidence*. An agent may reuse economic hypotheses, transformation rules, failure modes, and all market observations already revealed. It may not reuse the statistical evidence accumulated by a parent factor to certify a new descendant. Each candidate is therefore born as a *proof-carrying alpha*: an immutable expression coupled with a birth time, a predictable trading rule, a bounded net-utility definition, an evidence process, and an error budget. Only observations revealed after birth contribute to its certificate.

Figure 1 illustrates the resulting evidence firewall. The discovery ledger is deliberately adaptive: it stores factor lineages, outcomes, and distilled skills. The certification ledger is deliberately restrictive: it accepts an update only when the factor and its position were fixed before the corresponding return was observed. This prequential discipline lets us construct an e-process for every factor [7]. Because e-processes remain valid under optional stopping, the system can monitor candidates continuously. A predictable alpha-spending rule additionally controls the probability of ever certifying any null factor across an unbounded stream of adaptive proposals.

The framework also adapts the selection–utilization–distillation lifecycle of skill-augmented agents [11] to delayed financial outcomes. A skill represents an economic mechanism, an expression template, an applicable regime, and known failure boundaries. We use bounded increments of log evidence rather than raw e-wealth to credit the three capabilities. This choice supplies a stable learning signal while leaving certification to the unmodified e-process and its pre-registered threshold.

Our empirical design separates reliability from raw backtest performance. A synthetic null market tests whether false discoveries inflate with search depth. A planted-signal market measures power and time to certification. Finally, chronological A-share and U.S. equity panels test locked generalization and net portfolio value. Every method receives the same factor grammar, market data, LLM, and candidate budget. We report not only the best RankIC and Sharpe, but also family-wise error, false discovery rate, the search–generalization gap, certification delay, factor redundancy, and inference cost.

Our contributions are threefold:

- We formulate *adaptive discovery inflation* in agentic alpha mining and provide a null-market protocol that measures how reported validation performance changes with search depth.
- We propose ALPHA^2, a proof-carrying self-evolution architecture whose evidence firewall permits unrestricted learning from the past while preventing descendants from inheriting statistical evidence.
- We establish anytime-valid per-factor certification and stream-wide family-wise error control for adaptively generated factors, and define a reproducible evaluation of the reliability–performance frontier.

2 Related Work

2.1 Formulaic Alpha Mining

Early automated alpha discovery used genetic programming and symbolic search to construct interpretable expressions. Recent learning-based methods optimize the search distribution itself. AlphaGen uses reinforcement learning to generate a synergistic set of factors rather than isolated high-scoring expressions [21]. An earlier system named Alpha^2 uses deep reinforcement learning to construct logical formulaic alphas [18]; we distinguish it below as *Alpha*^{2}*-DRL*. AlphaForge combines a generative–predictive network with dynamic factor weighting [10]. Risk-seeking and grammar-guided variants improve exploration of sparse expression spaces [8]. These methods make the search more efficient, but their fitness functions remain repeated estimates on finite historical windows.

LLMs add economic priors and program-level reasoning. FAMA combines cross-sample selection with a chain of experience [5]; AlphaAgent uses originality, hypothesis alignment, and complexity regularization to counter alpha decay [13]; RD-Agent(Q) coordinates factor and model development in a data-centric loop [4]; and an LLM-guided MCTS navigates formula edits using backtest feedback [12]. AlphaPROBE represents the factor pool as a DAG and retrieves seeds using posterior potential [2]. These regularizers improve diversity or generalization empirically, but do not attach an error guarantee to a factor discovered after an adaptive number of trials.

2.2 Self-Evolving Financial Agents

Persistent memory and reusable skills have become central to agent evolution. Skill1 jointly trains skill selection, utilization, and distillation from a shared outcome signal [11]. FactorMiner instantiates a related loop for high-frequency alpha discovery, storing successful patterns and failure constraints in experience memory [16]. EvoQuant optimizes existing strategies through typed edits and a multi-stage verifier [6]. The last system is especially close in motivation, but its promotion gate is a collection of empirical thresholds; it does not provide optional-stopping or search-history-wide error control.

PACE reframes acceptance in self-evolving agents as an anytime-valid paired test [9]. It supplies a per-decision false-commit guarantee for prompt or workflow edits evaluated on benchmark instances. ALPHA^2 addresses a different statistical object: factors arrive at adaptive calendar times, their outcomes are dependent market returns, and the system must control false certification over the entire candidate stream. Our evidence firewall also formalizes what descendants may inherit when the agent learns from prior evaluations.

2.3 Data Snooping and Anytime-Valid Evidence

Repeated strategy search is a long-standing concern in financial economics. White's reality check tests whether the best model outperforms a benchmark after data snooping [17]; the deflated Sharpe ratio adjusts for selection and non-normality [1]; and the factor-zoo literature argues that new cross-sectional predictors require substantially higher significance thresholds [3]. These approaches primarily analyze a fixed or retrospectively enumerated family. Agentic mining instead creates new hypotheses conditional on earlier outcomes and may stop at arbitrary times.

E-values and e-processes provide evidence measures that remain valid under optional stopping and continuation [7]. E-BH controls FDR under arbitrary dependence among e-values [15], while e-backtesting applies sequential evidence to forecast calibration and financial risk monitoring [14]. We build on this theory but address a different problem: the hypothesis itself is generated by a learning agent. The key requirement is therefore not only an e-process, but a filtration-aware protocol that specifies when a factor is born, which

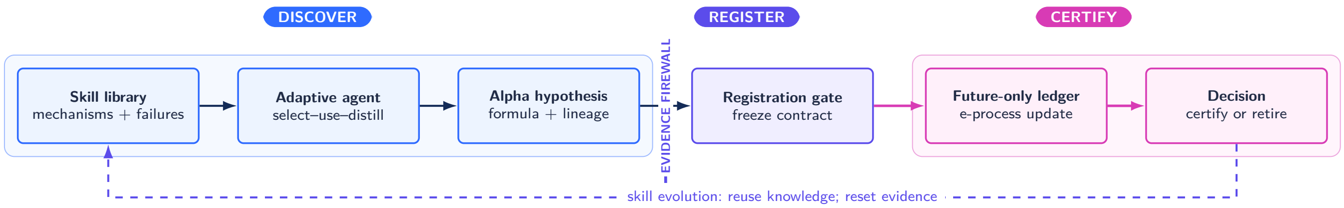

Figure 1: Overview of Alpha^2. The discovery side may adapt to all previously revealed outcomes. Registration freezes a candidate's formula and evidence contract. The certification side consumes only post-birth returns. Certified and failed lineages update the skill library, but every descendant restarts its evidence at one.

observations it may use, and how error budgets are spent across future descendants.

3 Problem Formulation

3.1 Adaptive Alpha Mining

Let $\{\mathcal{F}_t\}_{t\geq 0}$ be a filtration containing all market data, agent actions, factor evaluations, and library states revealed by trading date t. A stock panel at date t contains lagged features $X_t \in \mathbb{R}^{N_t\times d}$ and next-period returns $Y_{t+1} \in \mathbb{R}^{N_{t+1}}$. A formulaic factor is an executable map $f : \mathcal{X} \to \mathbb{R}^{N_t}$ assembled from a typed domain- specific language (DSL). Its signal is $z_t = f(X_{\leq t})$.

An adaptive miner maintains a policy π_{θ_t} and a library $\mathcal{B}_t$. At a data-dependent birth time τ_j, the policy proposes factor f_j using the complete history $\mathcal{F}_{\tau_j}$. Thus both f_j and τ_j are random, and the sequence of candidates cannot be treated as a fixed family selected before evaluation.

For each live factor, a portfolio map converts yesterday's signal into a predictable position $w_{j,t}$ satisfying $w_{j,t} \in \mathcal{F}_{t-1}$. The one-step net utility is

$$U_{j,t} = g_j\left(w_{j,t}^{\top} Y_t - c_j(w_{j,t}, w_{j,t-1})\right) \in [-1, 1], \tag{1}$$

where c_j is a pre-registered cost rule and g_j is a bounded monotone transformation. Bounding is required for the simple betting construction used below; it is not a claim that raw market returns are bounded.

Definition 1 (Net-positive alpha). *A registered factor j is a net-positive alpha over its certification regime if its predictable strategy has positive conditional net utility on a non-negligible set of post-birth dates. We test the composite null*

$$H_{0,j}: \quad \mathbb{E}[U_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \quad \text{for every } t > \tau_j. \tag{2}$$

This null is deliberately operational. Certification supports a claim about a fully specified, cost-adjusted trading rule, not merely positive cross-sectional correlation. RankIC and portfolio statistics remain useful descriptive metrics, but do not replace the evidence contract.

3.2 Adaptive Discovery Inflation

Suppose a miner evaluates M candidates on the same finite validation window and reports $\max_{j\leq M} \widehat{Q}_j$. Even when all candidates are null, the maximum increases with M under mild conditions. In an agentic system the problem is stronger: f_{j+1} is explicitly generated from the observed values of $\widehat{Q}_{1:j}$. We call the resulting divergence between selected validation quality and untouched future quality *adaptive discovery inflation*.

For search budget M, we measure

$$\begin{aligned} j^*(M) &= \arg\max_{j\leq M} \widehat{Q}_j^{\text{adapt}}, \\ \text{SGG}(M) &= \max_{j\leq M} \widehat{Q}_j^{\text{adapt}} - \widehat{Q}_{j^*(M)}^{\text{lock}}. \end{aligned} \tag{3}$$

where the locked window is not revealed until all search decisions are frozen. On a valid null generator, a reliable certification mechanism should maintain its nominal false-certification rate for every tested search depth.

3.3 Design Requirements

An evidence-safe miner should satisfy four requirements:

(1) **Adaptive proposal:** new factors may depend on every previously revealed outcome and the full evolving skill library.
(2) **Future-only certification:** no return observed before a factor's immutable registration may support its certificate.
(3) **Optional continuation:** factors may be monitored, retired, or certified at data-dependent times without invalidating their evidence.
(4) **Stream-wide accounting:** error control must cover a random and potentially unbounded number of candidate births.

4 Alpha^2

4.1 Proof-Carrying Alpha Records

The unit stored by Alpha^2 is not a bare expression. Candidate j is a versioned record

$$\mathcal{A}_j = (f_j, \tau_j, h_j, s_j, g_j, c_j, \alpha_j, E_{j,\cdot}, \mathcal{L}_j), \tag{4}$$

where h_j is the economic hypothesis, s_j is the originating skill, g_j and c_j define net utility, α_j is its error allocation, $E_{j,\cdot}$ is its evidence process, and $\mathcal{L}_j$ records parents, prompts, code, and data versions. The tuple is hashed at registration. Any change to the expression, lag, universe, normalization, cost, or utility function creates a child record with a new birth time and fresh evidence.

Table 1: Positioning against closely related adaptive alpha and agent systems.

Method	LLM/agent search	Persistent skill/graph	Chronological gate	Anytime-valid	Stream-wide control
AlphaAgent [13]	✓	–	empirical	–	–
Alpha^2-DRL [18]	✓	–	empirical	–	–
AlphaPROBE [2]	✓	graph	empirical	–	–
FactorMiner [16]	✓	skill + memory	empirical	–	–
EvoQuant [6]	✓	optimization memory	multi-stage	–	–
PACE [9]	✓	general agent state	paired future data	✓	per decision
ALPHA^2	✓	skill + lineage	evidence firewall	✓	**strong FWER**

This representation makes a certificate auditable. A reported factor can be replayed from a market snapshot, and its claim is scoped to the exact strategy whose returns generated the evidence. It also prevents a common ambiguity in factor mining: selecting a factor by RankIC and later presenting evidence for a differently tuned portfolio.

4.2 Evidence Firewall and Dual Ledgers

ALPHA^2 separates state into two ledgers.

Discovery ledger. The discovery ledger stores all registered formulas, evaluation trajectories, lineages, failure labels, semantic embeddings, and distilled skills. It is part of $\mathcal{F}_t$ and is intentionally available to the policy. An agent may, for example, learn that a volatility-normalized reversal template failed in high-turnover regimes and condition its next proposal on that observation.

Certification ledger. The certification ledger stores only immutable contracts and post-birth evidence updates. For $t \le \tau_j$, $E_{j,t} = 1$. The first admissible utility is computed from a position formed after registration and a subsequently revealed return. Evidence is never copied across records, including parent– child and semantically equivalent factors.

The firewall does not hide past returns from the agent; doing so would prevent learning. Instead, it constrains the interpretation of those returns. Past data may justify a new hypothesis, but cannot simultaneously serve as new evidence for that hypothesis.

4.3 Anytime-Valid Evidence Update

For a live candidate, choose a predictable betting fraction $\lambda_{j,t} \in [0, 1-\epsilon]$, where $\lambda_{j,t}$ may adapt to past utilities but must be $\mathcal{F}_{t-1}$-measurable. The evidence process is

$$E_{j,t} = E_{j,t-1}\big(1 + \lambda_{j,t} U_{j,t}\big), \qquad E_{j,\tau_j} = 1. \tag{5}$$

We evaluate constant, online-Newton-step, and mixture betting strategies in the experiments. The simple update in Equation (5) is sufficient for the validity statements; betting changes power, not the null guarantee.

Candidate j is certified at stopping time

$$T_j = \inf\{t > \tau_j : E_{j,t} \ge 1/\alpha_j\}. \tag{6}$$

A low evidence value is not interpreted as proof that a factor is useless. Candidates may instead be retired for compute or capacity reasons according to a policy fixed from $\mathcal{F}_{t-1}$.

4.4 Predictable Error-Budget Allocation

Let $A_0 = \alpha$ be the global family-wise budget. At birth j, the allocator chooses $\rho_j \in [0, 1]$ using only $\mathcal{F}_{\tau_j}$ and assigns

$$\alpha_j = \rho_j A_{j-1}, \qquad A_j = A_{j-1} - \alpha_j. \tag{7}$$

Hence $\sum_j \alpha_j \le \alpha$ pathwise. A conservative default uses $\alpha_j = 6\alpha/(\pi^2 j^2)$. Our adaptive allocator modulates the spending rate with pre-registration features such as semantic novelty, complexity, and parent-skill reliability, while enforcing a per-birth cap. It never uses post-registration outcomes of the candidate being funded.

For exploratory factor libraries where FDR rather than any-false-certificate control is preferred, stopped e-values can additionally be passed to e-BH at a fixed reporting time [15]. We keep strong FWER as the primary claim because it maps cleanly to admitting factors into a production candidate pool.

4.5 Evidence-Aware Skill Evolution

A skill is a structured object

$$s = (h, \mathcal{G}, r, \mathcal{C}^+, \mathcal{C}^-), \tag{8}$$

containing an economic mechanism h, an expression grammar/template $\mathcal{G}$, an applicability description r, successful construction patterns $\mathcal{C}^+$, and failure boundaries $\mathcal{C}^-$. This is more reusable than storing one factor formula as one skill.

Raw e-wealth is unsuitable as an RL reward because its scale grows multiplicatively and depends on factor age. We instead define a bounded per-date evidence gain

$$\delta_{j,t} = \mathrm{clip}(\log E_{j,t} - \log E_{j,t-1}, -b, b)\,. \tag{9}$$

For an evaluation horizon H_j, the utilization outcome is $R_j^{\mathrm{use}} = H_j^{-1} \sum_t \delta_{j,t}$, augmented with explicit complexity, turnover, and redundancy penalties. Following the temporal-credit intuition of Skill1 [11], we assign:

$$R_j^{\mathrm{select}} = \mathrm{EMA}\{R_k^{\mathrm{use}} : s_k = s_j\}, \tag{10}$$

$$R_j^{\mathrm{distill}} = R_j^{\mathrm{use}} - operatornamemedian\{R_k^{\mathrm{use}} : k \in \mathcal{N}(j)\}, \tag{11}$$

where $\mathcal{N}(j)$ is a pre-birth reference set of comparable factors. The policy can be optimized with GRPO or a bandit update. These learning rewards do not determine certification; they only influence future proposals.

4.6 End-to-End Procedure

At each discovery round, ALPHA^2 performs the following operations:

(1) retrieve skills and parent factors using the current task, regime description, semantic similarity, lineage novelty, and past evidence gains;
(2) generate executable candidates under the typed DSL, run static checks, and deduplicate equivalent abstract syntax trees;
(3) register each surviving record and allocate a new error budget;
(4) form lagged positions and update evidence as new market returns arrive;
(5) certify, continue, or retire candidates without changing their records;
(6) distill successful and failed construction rules into the discovery ledger, and use them only for factors born in later rounds.

The loop is asynchronous: factors born on different dates can accumulate evidence concurrently. This avoids the artificial assumption that all candidates are generated before a single validation period begins.

5 Validity Analysis

We state the guarantees under the operational null in Equation (2). Proofs are included for completeness because the filtration and candidate-birth conditions are central to the method.

PROPOSITION 1 (POST-BIRTH ANYTIME VALIDITY). *Let τ_j be a stopping time. Suppose the complete record defining f_j, g_j, and c_j is $\mathcal{F}_{\tau_j}$-measurable; $U_{j,t} \in [-1, 1]$; and $\lambda_{j,t} \in [0, 1-\epsilon]$ is $\mathcal{F}_{t-1}$-measurable. Under $H_{0,j}$, the process in Equation (5) is a nonnegative supermartingale. Therefore, for every $a \in (0, 1)$,*

$$\Pr_{H_{0,j}}\left(\sup_{t>\tau_j} E_{j,t} \geq 1/a \mid \mathcal{F}_{\tau_j}\right) \leq a. \tag{12}$$

PROOF. Nonnegativity follows from $U_{j,t} \geq -1$ and $\lambda_{j,t} < 1$. Under the null and conditioned on $\mathcal{F}_{t-1}$,

$$\begin{aligned}\mathbb{E}[E_{j,t} \mid \mathcal{F}_{t-1}] &= E_{j,t-1}\left(1 + \lambda_{j,t}\mathbb{E}[U_{j,t} \mid \mathcal{F}_{t-1}]\right) \\ &\leq E_{j,t-1}.\end{aligned}$$

The stopped, post-birth process is thus a nonnegative supermartingale starting at one. Ville's inequality gives Equation (12) [7]. □

The proposition permits the factor to be generated from arbitrary past agent interactions. It does not permit using pre-birth utilities inside $E_{j,t}$. That distinction is the mathematical role of the evidence firewall.

THEOREM 1 (STRONG STREAM-WIDE FWER CONTROL). *Consider a finite or countably infinite stream of adaptively generated candidates. Let each α_j be nonnegative and measurable at birth, and suppose $\sum_{j\geq 1} \alpha_j \leq \alpha$ almost surely. If every null candidate satisfies Proposition 1 and its null status is fixed by its registered record at birth, then*

$$\Pr(\exists j \in \mathcal{H}_0, \exists t > \tau_j : E_{j,t} \geq 1/\alpha_j) \leq \alpha, \tag{13}$$

where $\mathcal{H}_0$ is the random set of true nulls.

PROOF. Conditioning on the history at each birth and applying Proposition 1 gives a crossing probability at most α_j for every null candidate. A countable union bound and the tower property yield

$$\Pr(\text{any null crossing}) \leq \mathbb{E}\left[\sum_{j\in\mathcal{H}_0} \alpha_j\right] \leq \mathbb{E}\left[\sum_{j\geq 1} \alpha_j\right] \leq \alpha.$$

The argument does not require independence across factors or dates. □

COROLLARY 1 (ADAPTIVE MONITORING AND RETIREMENT). *The guarantee in Theorem 1 is unchanged if the agent chooses when to inspect, pause, or retire each candidate using the revealed history. It is also unchanged if skill updates alter future candidates, provided every new record restarts from $E = 1$ and uses only post-birth observations.*

PROPOSITION 2 (EVIDENCE INHERITANCE IS UNSAFE). *If a child factor is selected as a function of utilities already observed and its initial evidence is set to the selected parent's terminal wealth, or is recomputed on the same observations, Proposition 1 need not hold.*

PROOF SKETCH. Under a global null, generate many parents and select the one with the largest realized wealth. The selected maximum has expectation greater than one in general, even though every fixed parent's wealth has expectation at most one. Initializing a child from that selected value therefore violates the unit- expectation condition required of an e-value. Replaying the same observations after a data-dependent formula choice creates the same selection bias. □

What the theorem does not claim. The guarantee concerns false certification relative to the conditional net-utility null. It does not imply that true alphas have monotonically increasing wealth, that every useful alpha will be certified quickly, or that a certified historical effect will survive arbitrary market regime shifts. Evidence can decrease under the alternative, and conservative spending can reduce power. These limitations motivate the planted-signal and real-market experiments below.

6 Experimental Design

The experiments are designed around three research questions.

RQ1: Inflation. Does increasing adaptive search depth raise the best validation score on markets with no predictive signal, and how large is the resulting search–generalization gap?

RQ2: Reliability. Does ALPHA^2 control false certification under adaptive candidate generation while retaining useful power and certification speed when signals exist?

RQ3: Utility. On real markets, does evidence-safe evolution improve the reliability–performance frontier relative to search regularization, heuristic verification, and non-evolving statistical baselines?

6.1 Markets and Temporal Protocol

Synthetic null markets. We use two complementary generators. The *model null* simulates a multivariate volatility process with time-varying covariance and zero conditional mean, then derives OHLCV-like lagged features from the simulated path. The *permutation null* retains the real feature panel and applies a deranged block permutation to future-return vectors, preserving cross-sectional dependence and local return clusters while breaking their alignment with signals. The model null supports exact type-I interpretation; the permutation null tests robustness under realistic marginals.

Planted-signal markets. We inject one or more known DSL expressions into the conditional mean:

$$Y_{i,t+1} = \sum_{k=1}^{K} \beta_k \, \widetilde{f}_k(X_{i,\le t}) + \varepsilon_{i,t+1}, \tag{14}$$

where ε_t follows the covariance and volatility process estimated from the corresponding real panel. We vary signal strength, half-life, regime duration, factor correlation, and transaction costs. Because ground-truth lineages are known, we can measure discovery recall in addition to portfolio performance.

Real equity panels. The primary datasets are point-in-time CSI 500/CSI 1000 and S&P 500 panels in a Qlib-compatible format [19]. The target snapshot covers 2010–2025, subject to final source availability. We use 2010–2016 for warm-start feature and skill construction, 2017–2022 for prequential candidate birth and certification, and 2023–2025 as a locked test disclosed only after all policies, factors, and portfolio rules are frozen. Four rolling-origin replications shift these boundaries to assess regime sensitivity. Universe membership, splits, dividends, and delistings are reconstructed point in time; the exact data version and exclusions will be released with the artifact.

6.2 Factor Language and Backtest

All miners receive the same typed DSL. Terminals include lagged open, high, low, close, volume, amount, and returns. Operators include arithmetic, delay, delta, rolling mean/standard deviation, rank, correlation, covariance, extrema, quantile, and signed power. We reject expressions with invalid lookbacks, future references, divide-by-near-zero behavior, or more than a fixed number of AST nodes.

Signals are winsorized and cross-sectionally normalized using information available at date t. The primary factor portfolio is dollar-neutral: long the top quintile, short the bottom quintile, equal-weighted within each side, and executed with a one-day lag. We report 10 bp one-way cost and stress 0, 20, and 30 bp. Because A-share shorting is operationally constrained, long-only Top-k results are reported separately and long–short results are interpreted as factor diagnostics rather than directly deployable returns.

6.3 Baselines

We compare four families under matched candidate and compute budgets:

- **Symbolic/RL mining:** random grammar search, genetic programming, AlphaGen [21], and AlphaForge [10];
- **LLM alpha agents:** FAMA [5], AlphaAgent [13], RD-Agent(Q) [4], and LLM-MCTS [12];
- **Evolution/memory:** AlphaPROBE [2], FactorMiner [16], and an alpha-domain adaptation of Skill1 [11];
- **Verification:** fixed IC/IR gates, deflated Sharpe ratio, Bonferroni fixed-horizon tests, EvoQuant-style multi-stage promotion [6], and PACE-style per-candidate acceptance [9].

When an original method is not designed for continuous factor birth, we retain its proposer and evaluate both its native gate and a common chronological gate. This separates proposal quality from certification quality.

6.4 Budgets and Reproducibility

Search depth is varied over $M \in \{50, 100, 250, 500, 1000\}$ registered candidates. LLM methods use the same base model, temperature schedule, maximum tokens, and total inference budget. Each configuration is repeated with at least five agent seeds and 200 null-market replications; the final replication count will be set by a pre-experiment Monte Carlo power analysis. Prompt, factor, and evaluation caches are shared only when doing so does not leak an outcome across methods.

6.5 Metrics

Statistical reliability. We report empirical FWER, FDR, false certifications per run, calibration curves against nominal α, discovery precision/recall on planted markets, and median dates to certification. Confidence intervals use binomial or replication-level bootstrap intervals as appropriate.

Predictive and economic utility. Metrics include Pearson IC, RankIC, ICIR, annualized net return, Sharpe ratio, maximum drawdown, turnover, and downside deviation. We distinguish BestAdapt(M), chosen using the adaptive window, from the same factor's Locked(M) score and report SGG from Equation (3).

Search and agent quality. We measure executable-candidate rate, unique factors per 100 proposals, pairwise signal correlation, AST and semantic redundancy, certified factors per million tokens, and the fraction of distilled skills reused successfully.

6.6 Ablations and Stress Tests

The main ablations remove: (i) the evidence firewall by replaying the adaptive window; (ii) evidence reset by letting children inherit parent wealth; (iii) skill distillation; (iv) failure memory; (v) lineage-aware retrieval; (vi) adaptive spending; and (vii) transaction-cost penalties. Additional stress tests

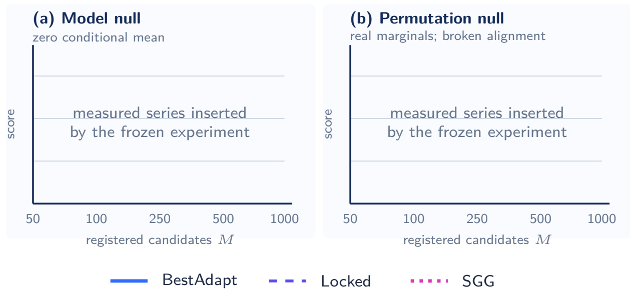

Figure 2: Result-ready template for adaptive discovery inflation. The frozen run will replace the labeled empty panels with measured BestAdapt, Locked, and SGG series and 95% replication intervals; no synthetic values are shown.

vary cost, missingness, universe churn, delisting treatment, volatility regime, search budget, LLM family, and betting strategy. The first two ablations are expected to improve apparent validation efficiency while violating calibration, which directly tests the mechanism behind our theory.

7 Results and Analysis

This section is intentionally structured before running the final experiments. Red placeholders specify the minimum result needed to support each claim; they must be replaced by measured values and uncertainty intervals before submission.

7.1 Adaptive Search Creates Alpha Illusions

Figure 2 will plot BestAdapt, Locked, and SGG against search depth on both null generators. The central diagnostic is whether BestAdapt rises with M while Locked remains centered at zero.

[RESULT: Report the slope of BestAdapt and Locked versus $\log M$, the SGG at $M = 1000$, and replication-level confidence intervals for every proposer.] This result establishes the phenomenon independently of our proposed method. If Locked also rises materially under the model null, the generator or evaluation pipeline is invalid and the downstream claims must not be made.

7.2 False-Certification Control and Power

Table 2 separates validity from power. A method is calibrated only if the upper confidence bound of empirical FWER is compatible with its nominal level. Search methods without multiplicity control are expected to deteriorate as M grows; this is an empirical hypothesis rather than an assumed result.

[RESULT: Insert FWER/FDR over at least 200 null replications, together with precision–recall and certification-delay curves over planted signal strengths.] The expected trade-off is conservative certification at weak signals and increasing

Table 2: Null and planted-signal reliability at $M = 1000$. Values are placeholders; FWER target is $\alpha = 0.05$.

Gate	Null FWER ↓	FDR ↓	Recall ↑	Delay ↓
IC/IR threshold	–	–	–	–
Fixed Bonferroni	–	–	–	–
PACE adaptation	–	–	–	–
ALPHA^2 fixed spend	–	–	–	–
ALPHA^2 adaptive spend	–	–	–	–

power as post-birth evidence accumulates. A failure to control null FWER would falsify either the implementation or the assumptions used in Theorem 1.

7.3 Locked Real-Market Performance

Table 3 reports the complete end-to-end systems. For each proposer, its native verification is compared against the common evidence-safe gate where computationally feasible. This avoids attributing all gains to the LLM when they arise from the acceptance rule.

[RESULT: Report locked metrics, paired seed-level tests against the strongest baseline, and multiplicity-adjusted uncertainty across markets.] The paper's main empirical claim should be framed as a reliability–performance frontier, not as universal dominance on every return metric. A strong outcome is that ALPHA^2 sharply reduces SGG and false promotion while remaining statistically indistinguishable from the best locked RankIC; higher Sharpe is desirable but not required by the theory.

7.4 What Makes Evidence-Safe Evolution Work?

The ablation table will report both Locked RankIC and null FWER. This paired view is necessary because an unsafe component can appear beneficial when judged only by selected validation performance.

[RESULT: Quantify the effects of evidence reset, future-only updates, skill distillation, failure memory, lineage retrieval, and adaptive spending. Include the number of candidate births consumed per certified factor.] We will also inspect representative lineages: one genuine planted factor recovered through successive skill refinements, one redundant family suppressed by failure memory, and one high-validation null factor rejected by the evidence ledger.

8 Discussion, Limitations, and Ethics

Certification versus deployment. A certificate rejects the registered conditional-mean null under the stated utility and cost contract. It is not a guarantee of future profit, market capacity, or robustness to an intervention that changes the data-generating process. Market impact, crowding, borrow availability, and structural breaks remain deployment questions. We therefore present certification as a stronger research filter, not as regulatory approval or investment advice.

Table 3: Locked 2023–2025 real-market results after costs. Values are reserved for the frozen experiment. Mean ± standard error will be reported over rolling origins and seeds.

System	CSI1000 RankIC	CSI500 RankIC	S&P500 RankIC	ICIR	Net Sharpe	Max DD	SGG	Cert./1M tok.
AlphaForge	–	–	–	–	–	–	–	–
AlphaAgent	–	–	–	–	–	–	–	–
RD-Agent(Q)	–	–	–	–	–	–	–	–
AlphaPROBE	–	–	–	–	–	–	–	–
FactorMiner	–	–	–	–	–	–	–	–
Skill1-Alpha	–	–	–	–	–	–	–	–
ALPHA^2	–	–	–	–	–	–	–	–

Power and delay. Future-only evidence prevents immediate certification of a newly generated factor. This delay is not an implementation defect; it is the price of not using the same realization for discovery and proof. The global FWER budget can also become conservative when thousands of candidates are proposed. Adaptive spending, e-BH reporting, hierarchical allocation by skill lineage, and more powerful mixture e-processes are promising extensions, but any improvement must preserve the filtration conditions used in the theory.

Non-stationarity. The null in Equation (2) is time-uniform and can be stringent. A factor that is useful only in a known, predictable regime should register the regime gate as part of its strategy. Retrospectively choosing the regime after observing returns would create a new hypothesis and require a new record. Certification may also be followed by decay; production monitoring should use a separate risk and calibration process rather than silently modifying the original factor.

Dependence and utility design. The validity proof permits arbitrary dependence across factors and dates but requires the conditional mean null and predictable bounded utility. Violations can arise from timestamp errors, same-close execution, revised fundamentals, or data-vendor survivorship. Our artifact therefore treats temporal-integrity tests as part of the certificate. The empirical FWER study is a necessary implementation audit, not a substitute for checking these assumptions.

LLM and benchmark contamination. LLMs may have memorized public alpha formulas, papers, or benchmark code. Formula novelty is consequently measured against both the starting library and known public factor sets. The main validity claim does not require candidates to be intellectually novel, but claims about discovery efficiency do. We will report model versions, prompts, temperatures, and inference dates, and repeat key comparisons with at least two model families.

Ethical scope. The study uses historical or simulated market data and does not execute live trades. Automated mining can contribute to crowding, speculative activity, and misleading performance claims. Releasing explicit temporal-integrity checks, cost assumptions, rejected candidates, and null-market results should reduce rather than amplify those risks. Any deployment remains subject to market-data licenses, financial regulation, internal model governance, and independent risk review.

9 Conclusion

Agentic alpha mining makes hypothesis generation cheaper, faster, and more adaptive. It also turns evaluation into part of the training loop, so a strong backtest can increasingly reflect search depth rather than predictive content. ALPHA^2 addresses this problem by allowing skills to learn from all revealed history while requiring every new factor to earn fresh, post-birth evidence. Its proof-carrying records, evidence firewall, e-process certificates, and predictable error spending provide a concrete separation between reusable knowledge and non-transferable statistical evidence. The theory establishes anytime-valid and stream-wide error control; the proposed experiments test the more practical question of whether this reliability can be achieved at useful power and locked market performance. More broadly, the framework suggests that the acceptor of a self-evolving scientific agent should be designed as carefully as its proposer.

A Implementation Contract

Every experimental run will persist the following fields for each candidate: run and seed identifiers; parent and skill identifiers; prompt and model version; canonical AST and executable hash; proposal and registration timestamps; feature, label, and universe snapshot hashes; maximum lookback; position lag; cost and utility definitions; allocated α_j; betting-rule parameters; daily utility and evidence; stopping reason; and all certification or retirement decisions. A candidate is invalidated if any field required to replay its signal is missing.

B Default Alpha-Spending Schedules

The deterministic baseline is

$$\alpha_j = \frac{6\alpha}{\pi^2 j^2}, \qquad \sum_{j=1}^{\infty} \alpha_j = \alpha.$$

The adaptive variant maintains remaining wealth A_{j-1} and chooses $\rho_j \leq \rho_{\max}$ from pre-registration covariates. To prevent a single early lineage from consuming the budget, the implementation reserves fixed fractions for top-level economic-mechanism groups and reports unused wealth. No allocator feature may contain the future outcome of the candidate receiving the allocation.

C Temporal-Integrity Tests

Before any alpha evaluation, the harness runs unit tests for: future-index access; same-bar execution; rolling-window boundary errors; inconsistent corporate-action adjustment; use of present-day constituents in historical universes; delisting omission; timezone mismatch; and cache keys that omit data cutoff time. We also run a label-shift test: moving labels one day into the future should destroy performance for lag-sensitive factors rather than leave it unchanged.

D Planned Statistical Reporting

All headline comparisons will include effect sizes and uncertainty intervals. Null-market error rates use exact binomial intervals. Real-market comparisons use paired replication-level bootstrap intervals across identical seeds and rolling origins. Hyperparameters are selected without access to the locked window. If multiple methods or markets are used to support one superiority claim, the associated tests will be corrected and the unadjusted values will be reported only as diagnostics.

E Artifact Checklist

The final artifact will include: the factor DSL; simulator and null generators; data-preparation manifests; baseline configurations; prompts; candidate and evidence ledgers; run scripts; environment lockfile; raw metric tables; figure generation notebooks; and a minimal replay script for every certified factor. Proprietary raw data, if any, will be replaced by a public-data reproduction and a schema-compatible loader.

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